Juan Sanchez Arias
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EDUCATION

Doctor of Philosophy | Neuroscience | GPA: 8.2/9.0 | University of Victoria | Victoria, BC 2015 - 2020
Doctor of Medicine | GPA: 4.4/5.0) | Universidad del Valle, School of Medicine | Cali, Colombia 2007 - 2014

RELEVANT CLINICAL EXPERIENCES

Observership: Functional Neurosurgery - 4 weeks | Supervisor: Dr. Konstantin Slavin 2014
- University of Illinois Hospital & Health Sciences System, Chicago IL

Pre-Diploma Internship (*Internado Rotatorio*) 2013 - 2014
- Hospital Universitario del Valle ESE | Universidad del Valle, School of Medicine | Cali, Colombia
  - **Internal Medicine (*Adulto y Anciano III*)** - 7 weeks
  - **Paediatrics (*Niño y Adolscente III*)** - 7 weeks
  - **Obstetrics and Gynaecology (*Salud de la Mujer III*)** - 7 weeks
  - **Specialized Research Internship in Biomedical Science (*Neuroscience Research*)** - 16 weeks
  - **Surgery (*Trauma y Violencia III*)** - 7 weeks
  - **Family Medicine (*Medicina Familiar*)** - 4 weeks

Professional Elective: Neurosurgery (*Neurocirugía*) - 4 weeks 2012
- Hospital Universitario del Valle ESE | Universidad del Valle, School of Medicine | Cali, Colombia

Professional Elective: Systems Pathology (*Patología de Sistemas*) - 4 weeks 2011
- Universidad del Valle, School of Medicine | Cali, Colombia

Professional Elective: Addiction and Pharmacodependency (*Adicción y Farmacodependencia*) - 1 semester 2008
- Universidad del Valle, School of Medicine | Cali, Colombia

SELECTED AWARDS

Research Trainee Award | **Michael Smith Health Research BC** 2021 - 2022
Nb Gulila Star Award | **International Gap Junction Conference** 2019
John & Myrtle Tilley Graduate Scholarship | **University of Victoria, Faculty of Graduate Studies** 2019
James & Laurette Agnew Memorial Scholarship | **University of Victoria, Faculty of Graduate Studies** 2015 - 2019
Donald Wagg Graduate Scholarship | **University of Victoria, Faculty of Graduate Studies** 2017 - 2018
Graduate Award | **University of Victoria, Faculty of Graduate Studies** 2015 - 2017
Fellowship Entrance Award | **University of Victoria, Faculty of Graduate Studies** 2015
School of Medicine Yearly Dean's List | **Universidad del Valle, School of Medicine** 2007, 2009, 2013, 2014

RESEARCH EXPERIENCE

Postdoctoral Research Fellow | University of Victoria, Division of Medical Sciences | Victoria, BC 2020 - 2022
- Led the design of a holistic and culturally safe translational research project to investigate an arrhythmia-causing genetic variant identified in the Gitxsan First Nation.
- Managed complex genomic and transcriptomic datasets using advanced bioinformatics (R, Bioconductor) to discover mechanisms underlying cardiomyocyte and neuronal calcium homeostasis.
- Communicated vital research outcomes via peer-reviewed journals and community-based talks, enhancing the collective knowledge of congenital cardiac conditions.

Graduate Research Assistant | University of Victoria, Division of Medical Sciences | Victoria, BC 2015 - 2020
- Established conditional knockout mouse models and optimized primary neuronal cultures to investigate mechanisms of dendritic spine plasticity and cerebral cortex development.
- Developed advanced microscopy methods to visualize cellular dynamics.

- Championed open-science practices by establishing accessible bio-image analysis frameworks for the research community.

Undergraduate Medical Researcher | Universidad del Valle, Centro de Estudios Cerebrales | Cali, Colombia 2014

- Facilitated literature seminars and thematic discussions for researchers and partners at the Centro de Estudios Cerebrales.
- Refined experimental procedures for modeling diffuse traumatic brain injury and developed staining techniques for organotypic brain tissue preparations.

Undergraduate Medical Researcher | Universidad del Valle, School of Public Health | Cali, Colombia 2010 - 2011

- Coordinated a regional survey assessing cardiovascular risk factors in spinal cord injury patients, identifying key social determinants of health to inform targeted rehabilitation plans.
- Collaborated with Physical Medicine and Rehabilitation residents and attendings to prescribe comprehensive, evidence-based management plans for chronic patients.

MEDICAL EDUCATION & TEACHING EXPERIENCE

Teaching Assistant | Island Medical Program, UBC | University of Victoria | Victoria BC 2017 - 2020

Foundations of Medical Practice I and II (MEDD412, MEDD421, MEDD422)

- Instructed first- and second-year medical trainees in neuroanatomy laboratory sessions, fostering a dynamic educational space centered on the functional integration of clinical signs with neural structures and physiological processes.

LEADERSHIP & COMMUNITY ENGAGEMENT

Builder/Hacker | **Placement: Third Place** | Island Health Code Hack | Victoria, BC 2023

- Collaborated within an interdisciplinary team to pitch and develop targeted technological solutions aimed at reducing "PJ paralysis" and preventing patient deconditioning across Island Health.

Co-creator & Organizer | "Who can become a scientist?" Symposium | Victoria, BC 2021 - 2022

- Established an educational symposium for high school students addressing equity, diversity, and inclusion in STEM, advocating for underrepresented populations in the sciences.

Local Chapter Coordinator | CIHR Brain Bee | University of Victoria | Victoria, BC 2017 - 2023

- Directed local operations and social media for the Victoria Brain Bee, mentoring high school students and securing funding to sponsor winners for national competitions.

Newsletter Co-editor | BC Regenerative Medicine Network| Victoria, BC 2017 - 2019

- Curated and edited organizational communications, recruited symposium speakers, and evaluated scientific abstracts to promote regional collaboration in regenerative medicine.

Student Representative | Neuroscience Graduate Program | University of Victoria| Victoria, BC 2017 - 2018

- Spearheaded collaborative programs dedicated to championing psychological wellness and mental health support for graduate trainees.
- Orchestrated the recruitment of distinguished researchers for the Neuroscience Graduate Program Seminar Series, facilitating meaningful student engagement with global experts.

WORK EXPERIENCE

Data Scientist | BenchSci | Toronto, ON (Remote) 2025 –Present

- Architecting AI-driven data intelligence pipelines and hardening extraction systems to deliver high-throughput API endpoints to support agentic preclinical biomedical acceleration.

Science Tech Lead and Team Lead | BenchSci | Toronto, ON (Remote) 2023 – 2025

- Led multidisciplinary teams and directed technical strategy to improve the observability and scientific quality of data pipelines.
- Engineered comprehensive analytics frameworks to evaluate disease relations across journal articles (MONDO, ICD-10, SNOMED), ensuring robust data quality metrics and standards.

- Partnered with cross-functional stakeholders (Product, Engineering) to align scientific requirements with shared technical execution and delivery.

Scientist II - Data Extraction | BenchSci | Toronto, ON (Remote) 2022 - 2023

- Developed and optimized biomedical data extraction workflows to enhance precision and data quality.
- Optimized preclinical research projects at top pharmaceutical companies using artificial intelligence.

PROFESSIONAL DEVELOPMENT AND CERTIFICATIONS

Basic Life Support - HCP, Airway Management, Oxygen Therapy, **Canadian Red Cross** 2026
Google Data Analytics Specialization, **Google Cloud** 2026
Certified Google Generative AI Leader, **Google Cloud** 2025
Certified Scrum Product Owner (1920762), **Scrum Alliance** 2024 - 2026
Course in Genetic Epidemiology and Epigenetics, **Universidad del Valle, School of Public Health** 2022
Code in Place , **Stanford University** 2021

PROFESSIONAL AFFILIATIONS

MEMBER | Colegio Colombiano de Neurociencias (COLNE) 2020 - Ongoing
MEMBER | Canadian Cardiovascular Society (CSS) 2021 - 2022
MEMBER | Canadian Association for Neuroscience (CAN) 2015 - 2022
MEMBER | Society for Neuroscience (SFN) 2016 - 2020

SELECTED PEER-REVIEWED PUBLICATIONS (Additional entries available in Google Scholar profile)

York, Nicole S; **Sanchez-Arias,** Juan C; McAdam, Alexa CH; Rivera, Joel E; Arbour, Laura T; Swayne, Leigh Anne. *Mechanisms underlying the role of ankyrin-B in cardiac and neurological health and disease*. Frontiers in Cardiovascular Medicine 9, 964675. **2022**. Frontiers Media SA.

Kallas, Dania; Bos, Johan Martijn; Peltenburg, Puck; Franciosi, Sonia; Mohajeri, Arezoo; Cheung, Matthew; Brett, Laura; York, Nicole; **Sanchez-Arias, Juan C**; Swayne, Leigh Anne. *PO-655-05 IS ANKYRIN-2 A POTENTIAL GENETIC MODIFIER IN PEDIATRIC CATECHOLAMINERGIC POLYMORPHIC VENTRICULAR TACHYCARDIA?*. Heart Rhythm 19,5,S264-S265. **2022**. Elsevier.

Sanchez-Arias, Juan C; Carrier, Micael; Frederiksen, Simona D; Shevtsova, Olga; McKee, Chloe; van der Slagt, Emma; Gonçalves de Andrade, Elisa; Nguyen, Hai Lam; Young, Penelope A; Tremblay, Marie-eve. *A systematic, open-science framework for quantification of cell-types in mouse brain sections using fluorescence microscopy*. Frontiers in Neuroanatomy. 15, 722443. **2021**. Frontiers Media SA.

Sanchez-Arias, Juan C; van der Slagt, Emma; Vecchiarelli, Haley A; Candlish, Rebecca C; York, Nicole; Young, Penelope A; Shevtsova, Olga; Juma, Afnan; Tremblay, Marie-Eve; Swayne, Leigh Anne. *Purinergic signaling in nervous system health and disease: Focus on pannexin 1*. Pharmacology & therapeutics 225, 107840. **2021**. Pergamon.

Sanchez-Arias, Juan C; Candlish, Rebecca C; van der Slagt, Emma; Swayne, Leigh Anne. *Pannexin 1 regulates dendritic protrusion dynamics in immature cortical neurons*. ENeuro,7,4,. **2020**. Society for Neuroscience.

Swayne, Leigh Anne; Johnstone, Scott R; Ng, Chen Seng; **Sanchez-Arias, Juan C**; Good, Miranda E; Penuela, Silvia; Lohman, Alexander W; Wolpe, Abigail G; Laubach, Victor E; Koval, Michael. *Consideration of Pannexin 1 channels in COVID-19 pathology and treatment*. American Journal of Physiology-Lung Cellular and Molecular Physiology 319,1,L121-L125. **2020**. American Physiological Society Bethesda, MD.

Sanchez-Arias, Juan C; Liu, Mei; Choi, Catherine SW; Ebert, Sarah N; Brown, Craig E; Swayne, Leigh Anne. *Pannexin 1 regulates network ensembles and dendritic spine development in cortical neurons*. Eneuro, 63. **2019**. Society for Neuroscience.

HOBBIES & INTERESTS

Sports: Strength conditioning, football, road cycling | **Technology**: Hackathons, Free Open Source Software Development, AI-driven Software Development

LANGUAGES

English: Fluent | **Spanish**: Native